1) 10000 Random Number Generated :-

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==========================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
==========================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 2
Enter dataset size (1-10000): 10000

Choose arrangement:
1. Sorted order (ascending)
2. Reverse-sorted order (descending)
3. Random order
Enter choice: 3
Dataset generated successfully.
```

```
   SORTING ALGORITHM PERFORMANCE ANALYZER
========================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
----------------------------------------
Enter your choice: 3

Original master dataset:
12291 20636 15577 29177 31043 29399 3446 18038 25609 18440 25239 2364 5250 3705 16618 11659 27903 11017 5734 21455
3400 31534 21564 19399 23603 2102 25866 486 20341 32722 21330 23546 26039 5588 13356 18699 1773 13112 10807 2182
16099 5729 5333 27803 1560 18598 16280 7139 3280 3341 9719 5702 15461 11350 8006 284 25310 16956 1765 32402
15554 28663 13495 5199 20783 6149 22997 25944 21542 1834 5508 17201 809 11958 22117 23110 12296 21510 23532 32137
28152 22861 27667 21851 5349 917 4432 30128 3849 25088 29892 8046 9406 14932 10421 12567 14992 7862 30526 2728
13719 4278 2373 26817 31236 21900 31150 31493 14329 24481 10446 14788 31390 29010 25635 16088 11633 23792 13890 9676
3037 1547 29456 14148 12047 2032 21557 25496 11781 9584 14798 20675 999 26083 25442 32636 12443 877 13592 19362
18324 3712 23847 19421 24810 27298 28045 21635 8368 2161 15876 8632 18562 228 10802 27866 1885 30163 32 28956
31667 23482 13037 9072 7103 4282 9668 8619 30743 25824 8971 7395 3200 30634 21469 17180 4869 3783 19387 27684
3403 18975 31870 3516 30751 22803 9872 15096 23889 5382 6492 13033 14277 9500 22531 17017 24007 3905 30289 32149
26015 23150 16879 5010 14437 9767 3034 17097 18199 23369 9747 12165 5609 23042 27033 8736 4572 18721 22676 28776
25746 27280 12499 5703 14945 6430 632 16804 11942 21378 7922 21892 21084 30877 29740 10357 237 6400 10244 556
30535 430 2047 7277 24883 25635 8507 11051 20706 16277 5285 12273 21705 28537 31595 7296 8112 841 28597 2234
30469 17285 3472 32702 23413 2040 42 10325 8126 17212 28485 19597 25022 26496 24806 1187 6608 2185 25421 2342
13401 24311 4906 28762 27038 4298 21006 23259 18393 25380 16672 21378 9099 14672 21521 26252 25107 1284 4178 13305
21120 3043 19588 307 2212 9798 1072 3568 28265 23356 5048 12377 32617 32208 28813 31852 25272 12170 12707 31391
22276 27704 31778 12742 9441 749 10335 2102 2910 8267 13066 16427 23260 19966 20137 4967 12832 2953 10564 12828
4518 25669 14946 10508 14648 9894 11573 29423 960 8385 15122 14613 13659 1547 2596 1868 28590 25188 25696 30468
16509 15956 23227 5553 32377 22676 10395 2673 26714 6352 27532 16111 14894 8107 821 3975 11505 6148 11658 18091
5213 11639 13123 26501 3843 19495 14476 3037 31899 3515 1534 22846 11929 30631 24014 20820 1793 28333 12612 409
31099 931 19055 22065 8874 32118 15896 5137 27066 17519 14836 15975 24632 4789 797 27446 21543 3604 14531 19265
18512 12028 12063 26923 18137 6349 30377 25564 16921 18257 25483 10279 30635 13706 10581 28073 2497 25007 24306 31476
17083 21105 32475 29933 14073 386 14218 15730 14573 20312 8242 9536 21364 25647 9645 29972 15974 8781 17334 4625
363 3465 24152 16738 14609 19470 20390 20256 12828 9309 32173 7286 20254 7313 3239 22787 27719 7911 370 27519
13693 18301 26688 19874 8879 16218 21391 24385 3687 18607 18268 29883 6815 24668 25747 9363 22486 6326 26827 6238
25187 4488 22064 25824 27144 12632 22464 11634 22474 1003 13755 5798 1536 25099 28244 19502 16043 18055 18093 7674
29701 29820 26626 30881 26461 21203 23845 15618 12833 8459 1066 15147 25917 26936 18531 21626 12963 21324 14479 2305
21803 15766 19790 29743 10910 16008 29656 31917 13325 29629 12527 15239 28776 16955 28690 24204 17740 27637 21236 2551
27163 7828 5420 32079 2386 24438 22574 32651 3095 30967 27514 22138 17426 19696 25875 19810 26494 12414 7731 3816
4992 13573 26685 22063 18615 628 3571 15172 5652 9930 29929 7866 26092 32473 29497 30865 8523 7241 11641 3677
7564 19681 4330 21667 2086 9986 15858 32647 17609 16244 30413 1700 29098 27750 26239 8252 1563 14767 21517 30609
16222 29274 5612 8850 20411 25442 22970 6842 976 15290 5164 15670 21033 11664 11592 5702 2320 13523 15572 31717
3750 9971 27406 15276 26419 28648 18845 24564 30139 8309 14536 20539 10962 12718 11494 20681 5552 8632 26645 9618
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6120 30558 1274 23840 749 13109 30720 14503 22992 23660 13309 6664 16606 9195 3273 4624 8388 15993 22428 1636
22693 7809 19909 22886 22122 31509 1950 18776 2394 18483 9151 19140 25322 1095 28274 25115 22998 32498 7527 16540
9072 32010 29609 10359 5944 24034 25041 23603 28846 9675 30330 7501 25444 4570 13044 32290 13655 6380 6205 2830
21375 32232 44 3798 5285 22340 5872 30275 7797 12235 2233 5904 8404 9373 14261 24071 23412 9335 3646 10342
8613 13364 16722 19591 20590 23700 5955 5568 27356 9533 1711 14217 20634 16639 16819 31708 17090 20316 348 12514
10995 22344 9566 6603 22692 24365 31753 26365 10423 17970 18040 31090 19281 17085 19724 30043 7146 10806 6169 26634
16288 26127 8414 1294 32209 14849 3771 16032 1768 16280 21662 15678 7543 15089 17087 31381 21895 11459 24717 932
24108 17820 8935 9759 2396 7789 21136 3345 1536 23113 14045 4294 76 5482 22705 15848 7244 17781 29099 22504
32448 28414 23454 23199 22570 21583 5186 16182 9765 357 10537 2115 7928 22844 8180 22836 31595 2125 11974 7409
4826 84 22777 32701 32135 24918 17424 16477 12596 29667 22765 7293 19442 6297 1349 23631 308 20713 12624 16757
23847 17694 15813 23841 21062 27546 7685 4475 17705 13667 25238 16963 2717 25567 18388 30903 3066 27612 7395 32439
1457 2515 5864 13607 5157 27502 25721 10645 700 4679 5458 12059 4405 21491 10825 32391 9020 6975 24964 8634
3643 20153 4841 11854 31092 115 12502 4988 1101 19203 16658 32572 9962 32718 25092 7853 257 2695 12323 15800
2794 11762 8110 2544 23798 32238 7837 31933 258 15026 4823 12474 588 29015 2213 9819 12165 1431 9575 15315
17191 2260 12934 23605 26120 603 21673 10713 16432 3307 17196 10241 3047 28017 5070 10172 24339 1814 28329 31755
7873 20603 15976 14659 20738 11843 12468 946 16632 22227 6895 12677 10544 9045 5925 17385 22578 29187 27967 31601
855 32162 7290 3000 15323 20044 6424 12871 14899 10099 4694 13497 29601 5923 21661 2589 11781 2695 7164 13211
1526 2991 27350 2139 6929 3216 14552 30882 22161 25421 3630 3134 33 23477 24311 8229 5358 524 15094 10073
5375 2830 9610 29454 15049 32049 10720 17935 20734 32555 32615 14174 6823 28596 21124 10266 370 10893 1416 20646
32333 7873 32223 10504 28723 6699 20873 12431 21830 28216 26831 22713 6687 20974 26670 5971 29538 9517 30360 25570
30101 31029 13276 9282 24370 617 29538 19597 18682 22355 3867 13887 12429 8423 26296 29335 4408 8204 24346 7232
32526 11352 30218 27696 30778 4123 15915 5585 24723 26016 22316 11870 23409 22959 13890 9104 30655 30550 21206 1403
25170 24810 2939 10162 4729 6728 22306 23986 9430 18528 5777 22142 22636 2549 6673 23430 25080 19870 12135 30413
2437 18705 19462 17483 17979 30112 10115 29329 19576 13828 23113 13995 16930 2394 21646 14154 3387 28523 16774 13091
28853 25136 10966 26258 27528 2198 28747 29827 3587 1662 26572 22162 14687 3592 14733 6257 32154 5649 19590 25226
2435 23034 22308 6118 24849 1707 26686 2894 19960 14552 31818 4016 15998 12419 29393 26564 1675 24003 30752 22766
14080 10947 14145 25959 32586 9997 26665 29080 19767 30729 3635 22726 7944 12992 31437 13556 15303 742 26737 24069
17883 17176 16856 25634 23513 27234 26986 2395 15261 29444 5368 32401 7066 7798 4525 26515 11886 15017 1090 27271
1209 19600 31090 20108 16688 17375 23214 28061 21229 7187 6929 32184 10761 3762 12438 30016 6765 15060 11657 3745
24805 26082 3643 12082 32375 14750 3834 13663 24050 9059 7744 29191 30665 14510 16171 20771 26658 17630 28717 30121
11051 25656 27141 19632 29613 20526 19182 23508 20451 16609 534 20188 4236 2020 3921 2202 9791 18499 27407 15123
26879 28612 10892 19168 32271 21868 13382 468 20487 8517 6324 21727 19752 24927 25397 16690 24557 28693 30535 1188
21986 3985 19961 13653 10425 24211 10147 31889 23963 2669 1492 30967 28348 25171 396 3034 29490 12241 20907 5607
30983 30877 5546 26376 17805 3471 14473 25993 28657 2798 8 12410 2579 11757 7194 16724 19349 9077 19095 20703
29491 13349 25393 850 21074 29080 24489 8124 14043 18662 13379 22180 10057 7789 3191 23532 9473 22225 16361 22524
26850 28993 5019 29724 19998 5884 7009 1568 24392 17428 20374 8011 14766 28018 11022 622 15080 18459 25931 17744
11441 5554 7412 8530 25334 6717 1666 14163 45 7980 29501 19154 21480 10135 17511 14418 1962 14791 7863 11011
3577 25080 29193 21918 1972 21085 10579 12945 5031 17509 5851 3932 15453 32573 28839 26427 30773 10134 28981 20741
28600 15528 28781 1108 32298 26230 2260 14822 8287 31060 7426 27459 17873 21159 20822 13728 14319 29421 19944 25604
2999 30973 23743 25769 20870 25231 14405 11685 30511 13822 7671 21705 4171 31053 31073 24266 30929 6761 15178 2462
1511 11437 12588 5673 14583 8107 28903 21785 10500 21290 11445 31503 5695 11371 865 32767 23509 4273 8372 3545
13324 26879 16945 17001 654 4900 29698 18556 6404 25589 6705 3140 10727 7898 6683 3837 23955 1676 14370 7633
15774 27420 32163 28665 27239 22840 15144 28710 20210 20138 31454 13599 18163 6177 30537 28075 27432 15412 27190 17329
30700 6052 18417 31390 522 3809 15033 19719 28111 30928 29297 19856 23651 9932 7320 20236 857 11755 6267 6117
8836 21890 15273 25790 3084 20203 15016 32344 29378 6789 2534 26290 29361 15869 19186 24949 2037 12236 6636 23438
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Selection Sort Result:-

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   SORTING ALGORITHM PERFORMANCE ANALYZER
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1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 4

Selection Sort result:
0 3 8 15 23 27 31 31 32 33 36 42 42 44 45 49 56 58 59 63
65 66 73 76 77 78 83 84 84 88 94 100 100 105 106 111 115 118 120 125
126 128 129 136 143 144 147 151 153 155 161 162 165 170 172 172 173 174 176 181
191 202 203 204 206 206 218 222 222 228 228 233 234 237 240 248 254 257 258 258
270 272 276 278 279 281 281 282 284 288 293 296 305 307 307 308 311 311 316 316
318 321 327 329 333 339 340 342 343 346 347 348 349 357 363 363 370 370 375 384
386 394 396 409 409 409 410 412 413 414 414 416 419 424 425 427 428 428 430 430
434 452 463 463 468 474 479 486 487 488 496 497 500 504 508 510 517 522 524 527
528 531 534 535 536 548 550 553 556 556 557 558 560 563 573 582 583 586 588 588
603 604 617 622 628 630 632 634 638 640 641 642 644 653 654 654 661 663 664 665
668 673 677 680 681 681 685 685 687 695 700 710 718 722 724 726 727 730 734 736
738 738 742 743 749 749 757 767 771 775 775 780 786 792 794 795 797 798 802 803
804 807 808 809 821 821 825 827 836 838 841 842 846 847 849 850 852 855 857 858
862 864 865 874 876 877 880 883 892 898 898 903 907 910 913 913 915 917 917 918
919 926 929 931 932 932 946 947 949 952 959 960 966 969 972 976 977 977 988 996
999 999 1000 1003 1005 1013 1024 1030 1030 1036 1037 1039 1042 1046 1052 1066 1067 1068 1072 1072
1073 1074 1074 1075 1078 1078 1079 1083 1090 1095 1095 1101 1103 1103 1103 1108 1108 1111 1112 1117
1120 1123 1129 1131 1147 1148 1154 1154 1159 1164 1181 1182 1187 1187 1188 1193 1195 1197 1205 1209
1209 1229 1234 1235 1238 1241 1251 1258 1259 1260 1269 1273 1274 1284 1287 1289 1291 1294 1296 1302
1311 1318 1318 1319 1324 1326 1326 1328 1334 1335 1343 1343 1349 1352 1355 1359 1363 1364 1369 1371
1378 1378 1392 1392 1402 1403 1410 1412 1414 1416 1417 1418 1419 1420 1421 1422 1425 1425 1431 1435
1435 1446 1457 1458 1459 1466 1466 1474 1480 1481 1482 1483 1488 1492 1493 1494 1497 1507 1508 1510
1511 1511 1512 1515 1520 1525 1526 1526 1534 1535 1536 1536 1543 1545 1547 1547 1547 1556 1557
1558 1559 1560 1562 1562 1563 1565 1567 1567 1568 1569 1570 1570 1570 1574 1577 1599 1611 1620 1631
1636 1636 1641 1643 1644 1654 1657 1662 1666 1668 1669 1671 1675 1676 1676 1677 1677 1678 1681 1684
1691 1692 1693 1695 1699 1699 1700 1700 1701 1707 1708 1711 1720 1723 1726 1730 1748 1754 1761 1765
1767 1768 1769 1771 1772 1773 1775 1780 1784 1784 1792 1793 1799 1808 1809 1812 1814 1814 1816 1827
1829 1831 1833 1834 1836 1837 1838 1838 1839 1839 1843 1847 1852 1858 1858 1864 1865 1868 1880 1880
1883 1885 1895 1905 1906 1909 1917 1920 1923 1936 1940 1941 1950 1950 1953 1957 1962 1963 1966 1967
1972 1972 1979 1988 1991 1991 2001 2001 2005 2007 2011 2012 2015 2020 2023 2026 2027 2029 2030 2032
2032 2033 2037 2039 2040 2045 2046 2047 2048 2060 2064 2065 2071 2079 2079 2081 2084 2085 2086 2086
2088 2094 2102 2102 2102 2107 2111 2115 2115 2119 2124 2125 2127 2139 2140 2145 2154 2155 2156 2161
```

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 2164 2164 2165 2170 2170 2173 2175 2182 2182 2185 2189 2191 2192 2195 2197 2197 2198 2198 2202 2211
 2212 2213 2216 2218 2219 2225 2226 2232 2233 2234 2239 2254 2260 2260 2261 2266 2266 2269 2270 2276
 2278 2278 2278 2283 2285 2289 2294 2298 2299 2303 2305 2306 2320 2324 2340 2340 2342 2345 2348 2348
 2348 2349 2352 2355 2355 2364 2372 2373 2376 2377 2379 2381 2386 2394 2394 2395 2395 2396 2396 2397
 2399 2401 2403 2411 2413 2413 2418 2419 2421 2425 2429 2430 2431 2435 2437 2441 2441 2444 2447 2451
 2453 2454 2460 2461 2462 2463 2464 2464 2467 2473 2476 2477 2480 2492 2494 2495 2497 2497 2509 2515
 2516 2518 2520 2520 2521 2524 2526 2528 2529 2529 2531 2534 2541 2541 2542 2542 2544 2545 2549 2549
 2551 2552 2553 2556 2557 2559 2560 2564 2565 2567 2571 2574 2579 2588 2589 2592 2593 2596 2597 2597
 2598 2622 2624 2626 2635 2648 2655 2660 2666 2669 2673 2674 2675 2678 2680 2681 2683 2684 2686 2690
 2691 2691 2692 2695 2695 2708 2709 2713 2715 2716 2717 2719 2720 2728 2728 2738 2738 2747 2750 2756
 2757 2770 2771 2774 2782 2783 2786 2787 2787 2790 2793 2794 2795 2798 2802 2812 2815 2819 2824 2828
 2830 2830 2830 2833 2833 2833 2837 2844 2849 2850 2852 2856 2868 2878 2880 2880 2881 2881 2882 2885
 2888 2894 2899 2900 2904 2906 2908 2908 2909 2910 2910 2910 2911 2915 2923 2936 2938 2939 2940 2942
 2942 2945 2947 2948 2950 2952 2953 2953 2955 2956 2958 2961 2970 2970 2971 2990 2991 2997 2997 2999
 3000 3006 3011 3019 3022 3024 3024 3032 3034 3034 3034 3036 3037 3037 3037 3038 3040 3041 3042 3043
 3047 3047 3052 3054 3057 3057 3058 3064 3066 3075 3076 3084 3095 3096 3100 3107 3109 3110 3113 3115
 3116 3119 3121 3122 3122 3124 3127 3128 3134 3134 3140 3143 3144 3146 3152 3156 3166 3167 3168 3168
 3168 3174 3177 3191 3194 3200 3204 3205 3205 3206 3215 3216 3231 3236 3239 3245 3251 3255 3255 3257
 3259 3261 3271 3273 3280 3282 3285 3290 3294 3296 3303 3307 3308 3316 3316 3321 3322 3331 3336 3337
 3341 3343 3345 3350 3351 3358 3359 3360 3368 3372 3373 3376 3378 3379 3381 3382 3382 3383 3385 3386
 3387 3387 3388 3395 3400 3402 3403 3405 3405 3406 3409 3416 3419 3426 3426 3429 3430 3430 3433 3438
 3440 3444 3446 3446 3448 3450 3458 3462 3463 3465 3466 3470 3471 3472 3473 3475 3476 3480 3485 3487
 3493 3502 3503 3506 3506 3511 3511 3512 3515 3516 3518 3518 3519 3521 3527 3533 3534 3534 3537 3543
 3545 3546 3551 3557 3561 3568 3569 3570 3571 3577 3577 3578 3587 3587 3592 3596 3603 3603 3604 3604
 3607 3607 3618 3620 3622 3629 3630 3632 3634 3635 3642 3643 3643 3646 3647 3653 3655 3656 3658 3659
 3666 3667 3669 3669 3674 3677 3685 3687 3688 3689 3694 3698 3698 3699 3705 3705 3707 3709 3709 3712
 3714 3716 3725 3727 3734 3734 3737 3738 3741 3744 3745 3750 3755 3758 3760 3762 3771 3776 3778 3779
 3780 3780 3780 3783 3787 3794 3797 3798 3799 3802 3803 3803 3806 3808 3808 3809 3814 3815 3816 3820
 3820 3831 3833 3834 3835 3837 3838 3843 3844 3845 3849 3850 3857 3857 3857 3859 3860 3861 3862 3865
 3867 3875 3882 3893 3894 3898 3899 3903 3904 3905 3909 3909 3913 3915 3921 3925 3929 3932 3933 3935
 3937 3942 3944 3946 3947 3947 3949 3952 3953 3960 3961 3965 3970 3975 3985 3986 3988 3995 3999 4006
 4008 4010 4016 4020 4023 4034 4040 4041 4043 4045 4045 4049 4054 4061 4061 4063 4064 4071 4073 4076
 4079 4088 4090 4104 4104 4104 4116 4121 4123 4125 4131 4142 4142 4158 4164 4171 4171 4176 4178 4178
 4180 4182 4182 4183 4185 4185 4186 4199 4203 4214 4220 4227 4229 4230 4231 4231 4236 4236 4240 4242
 4245 4246 4251 4252 4253 4255 4261 4263 4272 4273 4273 4278 4282 4286 4288 4288 4291 4291 4292 4294
 4298 4302 4303 4304 4305 4307 4316 4320 4326 4330 4330 4332 4334 4336 4340 4342 4343 4346 4347 4359
 4363 4375 4380 4381 4388 4390 4395 4399 4400 4401 4403 4405 4406 4408 4408 4423 4431 4432 4434 4441
 4442 4453 4459 4465 4469 4470 4470 4472 4473 4475 4475 4476 4477 4477 4483 4488 4489 4490 4494 4496
 4503 4508 4511 4514 4518 4524 4525 4525 4528 4529 4529 4543 4544 4550 4551 4560 4561 4562 4566 4567
 4567 4570 4571 4572 4572 4576 4577 4577 4589 4600 4604 4608 4609 4615 4622 4624 4624 4625 4629 4629
 4647 4650 4655 4659 4661 4662 4668 4671 4674 4679 4680 4689 4689 4694 4696 4702 4705 4716 4729 4737
 4737 4739 4744 4747 4747 4758 4763 4764 4770 4771 4773 4781 4786 4788 4789 4793 4801 4802 4812 4823
 4826 4826 4828 4829 4835 4836 4841 4849 4850 4859 4869 4874 4881 4883 4892 4897 4899 4900 4900 4901
 4906 4915 4925 4928 4933 4934 4934 4936 4938 4940 4942 4943 4960 4963 4967 4969 4974 4978 4988 4992
 5010 5012 5019 5022 5028 5028 5031 5035 5038 5039 5040 5043 5048 5049 5051 5052 5053 5054 5056 5058
 5062 5064 5069 5070 5070 5076 5076 5083 5086 5091 5092 5099 5104 5113 5116 5120 5134 5136 5136 5137
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30928 30929 30931 30933 30934 30938 30940 30943 30944 30945 30953 30954 30956 30957 30959 30962 30967 30967 30973 30980
30983 30989 30997 30999 31000 31002 31009 31011 31015 31023 31029 31035 31036 31038 31043 31048 31050 31051 31053 31055
31057 31059 31060 31063 31063 31065 31065 31069 31073 31075 31075 31075 31077 31081 31082 31084 31085 31090 31090 31092
31093 31096 31097 31099 31114 31114 31118 31119 31121 31124 31126 31126 31128 31139 31150 31151 31158 31162 31162 31170
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31241 31243 31245 31253 31259 31259 31263 31263 31273 31278 31278 31282 31287 31289 31294 31301 31302 31302 31304 31305
31306 31308 31311 31314 31321 31321 31325 31337 31344 31345 31355 31357 31361 31363 31364 31378 31379 31380 31381 31386
31390 31390 31390 31390 31391 31396 31396 31400 31400 31401 31405 31409 31411 31419 31420 31421 31421 31424 31427 31427
31437 31445 31449 31454 31457 31464 31472 31476 31478 31479 31487 31492 31493 31501 31503 31506 31509 31509 31517 31517
31518 31523 31524 31529 31534 31539 31547 31560 31567 31568 31571 31577 31578 31589 31590 31591 31594 31595 31595 31600
31601 31606 31607 31609 31614 31614 31617 31618 31619 31619 31620 31628 31629 31636 31636 31641 31645 31648 31651 31655
31656 31657 31667 31667 31669 31670 31670 31671 31677 31680 31680 31681 31681 31682 31682 31688 31694 31701 31702 31705
31707 31708 31709 31712 31714 31717 31725 31730 31735 31735 31737 31738 31743 31743 31753 31754 31755 31759 31765 31769
31769 31771 31778 31780 31788 31791 31791 31794 31798 31800 31803 31804 31810 31814 31818 31821 31821 31822 31824 31824
31827 31828 31828 31832 31843 31846 31852 31855 31855 31856 31860 31863 31869 31870 31873 31878 31882 31884 31889 31890
31895 31896 31899 31905 31914 31917 31920 31931 31933 31936 31943 31945 31946 31948 31949 31951 31958 31968 31973 31974
31978 31981 31983 31986 31986 31988 31990 31995 32000 32008 32009 32010 32012 32012 32022 32026 32034 32034 32037 32039
32039 32040 32042 32043 32047 32049 32053 32056 32057 32069 32079 32091 32107 32107 32111 32118 32120 32122 32129 32130
32133 32135 32137 32148 32149 32151 32154 32155 32159 32160 32162 32162 32163 32163 32164 32164 32166 32172 32172 32173
32173 32173 32176 32179 32181 32184 32184 32185 32194 32194 32202 32206 32208 32209 32209 32209 32212 32217 32223 32226
32231 32232 32234 32238 32239 32242 32243 32246 32247 32250 32252 32255 32255 32256 32260 32262 32271 32273 32280 32281
32290 32298 32298 32299 32302 32307 32308 32313 32319 32320 32321 32324 32329 32333 32337 32337 32343 32344 32345 32360
32360 32372 32375 32376 32377 32378 32381 32381 32382 32385 32388 32390 32391 32394 32395 32395 32399 32401 32401 32401
32402 32402 32406 32409 32413 32415 32422 32423 32427 32429 32433 32437 32439 32441 32441 32441 32448 32454 32456 32462
32462 32464 32465 32466 32466 32473 32475 32475 32476 32482 32482 32483 32483 32486 32487 32488 32491 32497 32498 32501
32508 32508 32510 32515 32517 32524 32526 32529 32539 32546 32547 32547 32550 32552 32553 32555 32557 32560 32562 32564
32570 32572 32572 32573 32576 32578 32580 32586 32593 32594 32597 32608 32612 32615 32616 32617 32620 32621 32622 32635
32636 32640 32645 32646 32647 32649 32651 32651 32657 32660 32662 32664 32666 32669 32672 32684 32687 32687 32693 32701
32702 32705 32712 32715 32718 32722 32722 32727 32732 32734 32736 32737 32737 32741 32743 32751 32755 32757 32765 32767

Comparisons: 49995000
Swaps:       9993
Time:        112.00000 ms
Sorted correctly: YES
```

Insertion sort result:-

```
    SORTING ALGORITHM PERFORMANCE ANALYZER
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1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 5

Insertion Sort result:
0 3 8 15 23 27 31 31 32 33 36 42 42 44 45 49 56 58 59 63
65 66 73 76 77 78 83 84 84 88 94 100 100 105 106 111 115 118 120 125
126 128 129 136 143 144 147 151 153 155 161 162 165 170 172 172 173 174 176 181
191 202 203 204 206 206 218 222 222 228 228 233 234 237 240 248 254 257 258 258
270 272 276 278 279 281 281 282 284 288 293 296 305 307 307 308 311 311 316 316
318 321 327 329 333 339 340 342 343 346 347 348 349 357 363 363 370 370 375 384
386 394 396 409 409 409 410 412 413 414 414 416 419 424 425 427 428 428 430 430
434 452 463 463 468 474 479 486 487 488 496 497 500 504 508 510 517 522 524 527
528 531 534 535 536 548 550 553 556 556 557 558 560 563 573 582 583 586 588 588
603 604 617 622 628 630 632 634 638 640 641 642 644 653 654 654 661 663 664 665
668 673 677 680 681 681 685 685 687 695 700 710 718 722 724 726 727 730 734 736
738 738 742 743 749 749 757 767 771 775 775 780 786 792 794 795 797 798 802 803
804 807 808 809 821 821 825 827 836 838 841 842 846 847 849 850 852 855 857 858
862 864 865 874 876 877 880 883 892 898 898 903 907 910 913 913 915 917 917 918
919 926 929 931 932 932 946 947 949 952 959 960 966 969 972 976 977 977 988 996
999 999 1000 1003 1005 1013 1024 1030 1030 1036 1037 1039 1042 1046 1052 1066 1067 1068 1072 1072
1073 1074 1074 1075 1078 1078 1079 1083 1090 1095 1095 1101 1103 1103 1103 1108 1108 1111 1112 1117
1120 1123 1129 1131 1147 1148 1154 1154 1159 1164 1181 1182 1187 1187 1188 1193 1195 1197 1205 1209
1209 1229 1234 1235 1238 1241 1251 1258 1259 1260 1269 1273 1274 1284 1287 1289 1291 1294 1296 1302
1311 1318 1318 1319 1324 1326 1326 1328 1334 1335 1343 1343 1349 1352 1355 1359 1363 1364 1369 1371
1378 1378 1392 1392 1402 1403 1410 1412 1414 1416 1417 1418 1419 1420 1421 1422 1425 1425 1431 1435
1435 1446 1457 1458 1459 1466 1466 1474 1480 1481 1482 1483 1488 1492 1493 1494 1497 1507 1508 1510
1511 1511 1512 1515 1520 1525 1526 1526 1526 1534 1535 1536 1536 1543 1545 1547 1547 1547 1556 1557
1558 1559 1560 1562 1562 1563 1565 1567 1567 1568 1569 1570 1570 1570 1574 1577 1599 1611 1620 1631
1636 1636 1641 1643 1644 1654 1657 1662 1666 1668 1669 1671 1675 1676 1676 1677 1677 1678 1681 1684
1691 1692 1693 1695 1699 1699 1700 1700 1701 1707 1708 1711 1720 1723 1726 1730 1748 1754 1761 1765
1767 1768 1769 1771 1772 1773 1775 1780 1784 1784 1792 1793 1799 1808 1809 1812 1814 1814 1816 1827
1829 1831 1833 1834 1836 1837 1838 1838 1839 1839 1843 1847 1852 1858 1858 1864 1865 1868 1880 1880
1883 1885 1895 1905 1906 1909 1917 1920 1923 1936 1940 1941 1950 1950 1953 1957 1962 1963 1966 1967
1972 1972 1979 1988 1991 1991 2001 2001 2005 2007 2011 2012 2015 2020 2023 2026 2027 2029 2030 2032
2032 2033 2037 2039 2040 2045 2046 2047 2048 2060 2064 2065 2071 2079 2079 2081 2084 2085 2086 2086
2088 2094 2102 2102 2102 2107 2111 2115 2115 2119 2124 2125 2127 2139 2140 2145 2154 2155 2156 2161
2164 2164 2165 2170 2170 2173 2175 2182 2182 2185 2189 2191 2192 2195 2197 2197 2198 2198 2202 2211
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11211 11213 11220 11221 11225 11236 11239 11241 11241 11241 11245 11245 11247 11248 11261 11265 11266 11266 11273 11284
11291 11293 11295 11296 11298 11299 11315 11317 11318 11324 11326 11336 11343 11349 11350 11350 11352 11353 11363 11368
11370 11371 11379 11381 11383 11384 11384 11403 11404 11405 11407 11411 11415 11417 11421 11422 11437 11441 11442 11443
11445 11447 11453 11453 11453 11456 11459 11464 11466 11466 11468 11475 11482 11482 11484 11486 11490 11491 11494 11503 11505
11508 11508 11509 11515 11517 11525 11529 11530 11530 11532 11534 11538 11549 11552 11555 11557 11566 11568 11568 11573
11580 11591 11592 11595 11595 11597 11599 11600 11603 11603 11603 11606 11608 11609 11616 11616 11626 11628 11628 11629
11630 11632 11633 11634 11634 11636 11637 11639 11641 11651 11653 11657 11658 11659 11662 11664 11667 11670 11681 11685
11691 11691 11696 11701 11707 11710 11712 11716 11717 11722 11722 11722 11725 11725 11728 11729 11733 11734 11738 11740
11749 11751 11751 11752 11753 11755 11757 11762 11765 11767 11772 11773 11775 11779 11780 11781 11781 11786 11788 11796
11798 11817 11820 11826 11828 11837 11841 11843 11852 11852 11854 11860 11866 11869 11870 11871 11872 11873 11883 11883
11886 11886 11894 11895 11897 11898 11899 11905 11907 11911 11914 11917 11924 11929 11935 11938 11942 11950 11951 11957
11958 11961 11966 11966 11970 11974 11977 11983 11985 11985 11989 11993 11997 12000 12001 12008 12013 12014 12014 12015
12020 12022 12028 12030 12035 12036 12038 12040 12040 12042 12044 12045 12047 12050 12050 12050 12054 12056 12059 12060
12063 12063 12065 12065 12069 12077 12078 12080 12082 12088 12093 12096 12113 12114 12121 12121 12124 12131 12133 12133
12135 12139 12156 12160 12160 12165 12165 12165 12165 12170 12170 12172 12181 12188 12195 12196 12200 12204 12205 12207
12214 12217 12217 12220 12225 12227 12228 12235 12236 12236 12241 12241 12243 12257 12257 12268 12271 12273 12273 12276
12291 12292 12292 12292 12293 12296 12300 12312 12313 12318 12323 12329 12341 12344 12348 12351 12358 12359 12373 12374
12377 12391 12392 12397 12398 12399 12399 12409 12409 12410 12411 12414 12414 12419 12425 12428 12428 12429 12431 12431
12438 12439 12443 12444 12453 12454 12456 12459 12462 12468 12474 12474 12478 12478 12483 12486 12493 12494 12494 12495
12496 12497 12497 12499 12499 12500 12500 12501 12502 12503 12512 12514 12515 12523 12523 12527 12529 12544 12546 12549
12550 12562 12563 12564 12565 12566 12566 12567 12571 12580 12580 12583 12586 12588 12588 12596 12603 12606 12606 12610
12610 12611 12612 12616 12618 12620 12623 12624 12625 12625 12632 12632 12636 12638 12657 12657 12659 12659 12661 12665
12668 12669 12671 12672 12674 12677 12677 12687 12688 12691 12696 12706 12707 12714 12718 12719 12720 12722 12725 12734
12735 12738 12739 12742 12757 12766 12773 12778 12782 12783 12788 12790 12793 12796 12796 12799 12800 12813 12819 12821
12823 12826 12828 12828 12830 12832 12833 12838 12838 12847 12855 12856 12858 12861 12867 12870 12871 12871 12873 12880
12880 12880 12881 12890 12892 12894 12909 12919 12919 12920 12925 12934 12936 12938 12938 12939 12945 12957 12963 12969
12974 12981 12983 12988 12989 12990 12992 12995 12998 13002 13005 13007 13024 13026 13033 13033 13037 13044 13045 13045
13057 13066 13072 13075 13076 13076 13077 13080 13081 13087 13089 13091 13093 13094 13095 13095 13097 13104 13107 13108
13109 13109 13109 13112 13120 13121 13123 13131 13138 13141 13154 13170 13170 13174 13189 13194 13196 13196 13197 13200
13204 13206 13211 13215 13219 13231 13233 13234 13244 13245 13248 13250 13253 13253 13254 13258 13259 13264 13268 13268
13275 13276 13281 13284 13284 13287 13294 13295 13297 13297 13303 13305 13309 13320 13322 13324 13325 13326 13330 13333
13334 13336 13337 13346 13349 13356 13364 13365 13366 13369 13370 13372 13375 13375 13377 13379 13380 13380 13381 13381
13382 13382 13383 13384 13386 13387 13387 13395 13401 13407 13412 13413 13417 13418 13419 13434 13440 13444 13447 13449
13450 13451 13452 13456 13470 13472 13476 13481 13488 13495 13497 13498 13503 13505 13509 13511 13513 13522 13523 13524
13533 13541 13545 13556 13558 13558 13567 13571 13573 13575 13576 13577 13592 13593 13593 13593 13595 13599 13602 13607
13619 13620 13622 13622 13623 13623 13624 13627 13627 13630 13630 13634 13636 13637 13638 13640 13650 13651 13652 13652
13653 13655 13656 13657 13657 13659 13659 13663 13667 13673 13673 13675 13679 13679 13684 13689 13690 13692 13693 13695
13696 13697 13700 13702 13706 13711 13712 13718 13719 13720 13725 13726 13728 13728 13729 13730 13730 13733 13736 13738
13745 13746 13748 13749 13751 13753 13755 13760 13761 13761 13763 13764 13774 13777 13790 13790 13793 13795 13801 13801
13803 13805 13815 13817 13818 13821 13821 13821 13822 13823 13828 13829 13834 13837 13839 13840 13843 13845 13846 13846
13848 13858 13859 13863 13868 13871 13877 13878 13882 13882 13887 13887 13890 13890 13891 13899 13900 13909 13919 13923
13923 13926 13935 13937 13939 13947 13955 13956 13956 13959 13963 13968 13969 13972 13972 13973 13976 13981 13984 13986
13995 13996 13997 14002 14002 14006 14010 14013 14017 14021 14023 14026 14030 14032 14036 14036 14037 14040 14043 14043
14044 14045 14049 14051 14051 14057 14062 14069 14073 14080 14082 14083 14096 14099 14103 14104 14105 14112 14121 14134
14136 14140 14143 14145 14148 14148 14148 14151 14154 14154 14160 14163 14164 14166 14172 14173 14174 14174 14176 14178
14182 14185 14189 14191 14191 14196 14197 14198 14200 14204 14210 14211 14214 14217 14218 14218 14219 14220 14221 14221
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30222 30223 30231 30233 30240 30242 30244 30245 30246 30247 30248 30248 30257 30258 30267 30269 30275 30288 30289 30295
30301 30301 30304 30310 30317 30330 30331 30333 30337 30342 30344 30347 30350 30351 30352 30357 30358 30360 30361 30363
30365 30374 30375 30377 30384 30384 30389 30406 30407 30409 30413 30413 30415 30415 30420 30437 30437 30439 30451 30454
30456 30467 30468 30469 30469 30475 30477 30478 30478 30479 30482 30485 30488 30502 30503 30509 30511 30512 30513 30519
30521 30526 30526 30529 30529 30535 30535 30537 30540 30544 30549 30550 30550 30551 30557 30558 30559 30559 30561 30568
30569 30576 30579 30582 30582 30585 30596 30601 30601 30603 30607 30609 30609 30616 30618 30618 30627 30629 30630 30631
30634 30635 30635 30636 30638 30638 30642 30642 30646 30655 30658 30664 30665 30680 30681 30683 30684 30695 30699 30700
30708 30720 30720 30723 30726 30727 30729 30729 30729 30734 30738 30743 30748 30749 30751 30751 30752 30759 30767 30768
30769 30770 30771 30773 30778 30778 30780 30782 30785 30786 30789 30804 30817 30819 30827 30836 30837 30846 30861 30863
30864 30865 30868 30876 30877 30877 30877 30879 30879 30881 30882 30885 30895 30900 30900 30902 30903 30912 30913 30916
30928 30929 30931 30933 30934 30938 30940 30943 30944 30945 30953 30954 30956 30957 30959 30962 30967 30967 30973 30980
30983 30989 30997 30999 31000 31002 31009 31011 31015 31023 31029 31035 31036 31038 31043 31048 31050 31051 31053 31055
31057 31059 31060 31063 31063 31065 31065 31069 31073 31075 31075 31075 31077 31081 31082 31084 31085 31090 31090 31092
31093 31096 31097 31099 31114 31114 31118 31119 31121 31124 31126 31126 31128 31139 31150 31151 31158 31162 31162 31170
31173 31173 31175 31187 31197 31201 31203 31207 31210 31217 31219 31229 31229 31230 31233 31233 31233 31236 31237 31240
31241 31243 31245 31253 31259 31259 31263 31263 31273 31278 31278 31282 31287 31289 31294 31301 31302 31302 31304 31305
31306 31308 31311 31314 31321 31321 31325 31337 31344 31345 31355 31357 31361 31363 31364 31378 31379 31380 31381 31386
31390 31390 31390 31390 31391 31396 31396 31400 31400 31401 31405 31409 31411 31419 31420 31421 31421 31424 31427 31427
31437 31445 31449 31454 31457 31464 31472 31476 31478 31479 31487 31492 31493 31501 31503 31506 31509 31509 31517 31517
31518 31523 31524 31529 31534 31539 31547 31560 31567 31568 31571 31577 31578 31589 31590 31591 31594 31595 31595 31600
31601 31606 31607 31609 31614 31614 31617 31618 31619 31619 31620 31628 31629 31636 31636 31641 31645 31648 31651 31655
31656 31657 31667 31667 31669 31670 31670 31671 31677 31680 31680 31681 31681 31682 31682 31688 31694 31701 31702 31705
31707 31708 31709 31712 31714 31717 31725 31730 31735 31735 31737 31738 31743 31743 31753 31754 31755 31759 31765 31769
31769 31771 31778 31780 31788 31791 31791 31794 31798 31800 31803 31804 31810 31814 31818 31821 31821 31822 31824 31824
31827 31828 31828 31832 31843 31846 31852 31855 31855 31856 31860 31863 31869 31870 31873 31878 31882 31884 31889 31890
31895 31896 31899 31905 31914 31917 31920 31931 31933 31936 31943 31945 31946 31948 31949 31951 31958 31968 31973 31974
31978 31981 31983 31986 31986 31988 31990 31995 32000 32008 32009 32010 32012 32012 32022 32026 32034 32034 32037 32039
32039 32040 32042 32043 32047 32049 32053 32056 32057 32069 32079 32091 32107 32107 32111 32118 32120 32122 32129 32130
32133 32135 32137 32148 32149 32151 32154 32155 32159 32160 32162 32162 32163 32163 32164 32164 32166 32172 32172 32173
32173 32173 32176 32179 32181 32184 32184 32185 32194 32194 32202 32206 32208 32209 32209 32209 32212 32217 32223 32226
32231 32232 32234 32238 32239 32242 32243 32246 32247 32250 32252 32255 32255 32256 32260 32262 32271 32273 32280 32281
32290 32298 32298 32299 32302 32307 32308 32313 32319 32320 32321 32324 32329 32333 32337 32337 32343 32344 32345 32360
32360 32372 32375 32376 32377 32378 32381 32381 32382 32385 32388 32390 32391 32394 32395 32395 32399 32401 32401 32401
32402 32402 32406 32409 32413 32415 32422 32423 32427 32429 32433 32437 32439 32441 32441 32441 32448 32454 32456 32462
32462 32464 32465 32466 32466 32473 32475 32475 32476 32482 32482 32483 32483 32486 32487 32488 32491 32497 32498 32501
32508 32508 32510 32515 32517 32524 32526 32529 32539 32546 32547 32547 32550 32552 32553 32555 32557 32560 32562 32564
32570 32572 32572 32573 32576 32578 32580 32586 32593 32594 32597 32608 32612 32615 32616 32617 32620 32621 32622 32635
32636 32640 32645 32646 32647 32649 32651 32651 32657 32660 32662 32664 32666 32669 32672 32684 32687 32687 32693 32701
32702 32705 32712 32715 32718 32722 32722 32727 32732 32734 32736 32737 32737 32741 32743 32751 32755 32757 32765 32767

Comparisons: 25010684
Shifts:      25000695
Time:        94.00000 ms
Sorted correctly: YES

----------------------------------------
```

| Algorithm | Comparisons | Swaps/Shifts | Average time (ms) |
|---|---|---|---|
| Selection Sort | 49995000 | 9993 | 117.000 |
| Insertion Sort | 25010684 | 25000695 | 73.000 |

```
Selection output is sorted: YES
Insertion output is sorted: YES
Both outputs are identical: YES

Interpretation:
- Selection Sort performs n(n-1)/2 comparisons for every arrangement.
- Insertion Sort usually performs fewer comparisons on already sorted data.
- Timing is averaged across the requested trials; very small runs may show 0.000 ms.
- Repeat with different sizes and arrangements to compare practical behaviour.
```

2) Manually Dataset

```
============================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
============================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
--------------------------------------------
Enter your choice: 1
Enter dataset size (1-10000): 10
Enter 10 integer values:
23
464
768
34
78
345
876
345
567
345
Dataset saved as the untouched master copy.

============================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
============================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
--------------------------------------------
Enter your choice: 4

Selection Sort result:
23 34 78 345 345 345 464 567 768 876
Comparisons: 45
Swaps:       7
Time:        0.00000 ms
Sorted correctly: YES

--------------------------------------------
```

```
============================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
============================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
--------------------------------------------
Enter your choice: 5

Insertion Sort result:
23 34 78 345 345 345 464 567 768 876
Comparisons: 24
Shifts:      15
Time:        0.00000 ms
Sorted correctly: YES

============================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
============================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
--------------------------------------------
Enter your choice: 6
Enter number of timing trials (1-100): 1

========== PERFORMANCE COMPARISON ==========
Dataset size: 10 | Timing trials: 1
Original dataset:
23 464 768 34 78 345 876 345 567 345

Algorithm          | Comparisons  | Swaps/Shifts | Average time (ms)
-------------------+--------------+--------------+------------------
Selection Sort     | 45           | 7            | 0.000
Insertion Sort     | 24           | 15           | 0.000

Selection Sort output:
23 34 78 345 345 345 464 567 768 876
Insertion Sort output:
23 34 78 345 345 345 464 567 768 876
Selection output is sorted: YES
```

```
Selection Sort output:
23 34 78 345 345 345 464 567 768 876
Insertion Sort output:
23 34 78 345 345 345 464 567 768 876
Selection output is sorted: YES
Insertion output is sorted: YES
Both outputs are identical: YES

Interpretation:
- Selection Sort performs n(n-1)/2 comparisons for every arrangement.
- Insertion Sort usually performs fewer comparisons on already sorted data.
- Timing is averaged across the requested trials; very small runs may show 0.000 ms.
- Repeat with different sizes and arrangements to compare practical behaviour.
```

3) Reverse-sorted order (descending)

```
==========================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
==========================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 2
Enter dataset size (1-10000): 500

Choose arrangement:
1. Sorted order (ascending)
2. Reverse-sorted order (descending)
3. Random order
Enter choice: 2
Dataset generated successfully.

==========================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
==========================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 3

Original master dataset:
500 499 498 497 496 495 494 493 492 491 490 489 488 487 486 485 484 483 482 481
480 479 478 477 476 475 474 473 472 471 470 469 468 467 466 465 464 463 462 461
460 459 458 457 456 455 454 453 452 451 450 449 448 447 446 445 444 443 442 441
440 439 438 437 436 435 434 433 432 431 430 429 428 427 426 425 424 423 422 421
420 419 418 417 416 415 414 413 412 411 410 409 408 407 406 405 404 403 402 401
400 399 398 397 396 395 394 393 392 391 390 389 388 387 386 385 384 383 382 381
380 379 378 377 376 375 374 373 372 371 370 369 368 367 366 365 364 363 362 361
360 359 358 357 356 355 354 353 352 351 350 349 348 347 346 345 344 343 342 341
340 339 338 337 336 335 334 333 332 331 330 329 328 327 326 325 324 323 322 321
320 319 318 317 316 315 314 313 312 311 310 309 308 307 306 305 304 303 302 301
300 299 298 297 296 295 294 293 292 291 290 289 288 287 286 285 284 283 282 281
280 279 278 277 276 275 274 273 272 271 270 269 268 267 266 265 264 263 262 261
260 259 258 257 256 255 254 253 252 251 250 249 248 247 246 245 244 243 242 241
240 239 238 237 236 235 234 233 232 231 230 229 228 227 226 225 224 223 222 221
```

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220 219 218 217 216 215 214 213 212 211 210 209 208 207 206 205 204 203 202 201
200 199 198 197 196 195 194 193 192 191 190 189 188 187 186 185 184 183 182 181
180 179 178 177 176 175 174 173 172 171 170 169 168 167 166 165 164 163 162 161
160 159 158 157 156 155 154 153 152 151 150 149 148 147 146 145 144 143 142 141
140 139 138 137 136 135 134 133 132 131 130 129 128 127 126 125 124 123 122 121
120 119 118 117 116 115 114 113 112 111 110 109 108 107 106 105 104 103 102 101
100 99 98 97 96 95 94 93 92 91 90 89 88 87 86 85 84 83 82 81
80 79 78 77 76 75 74 73 72 71 70 69 68 67 66 65 64 63 62 61
60 59 58 57 56 55 54 53 52 51 50 49 48 47 46 45 44 43 42 41
40 39 38 37 36 35 34 33 32 31 30 29 28 27 26 25 24 23 22 21
20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1


==========================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
==========================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 4

Selection Sort result:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120
121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140
141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160
161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180
181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200
201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220
221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240
241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260
261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280
281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320
321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340
341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360
361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380
381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400
401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420
421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440
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361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380
381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400
401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420
421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440
441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460
461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480
481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500

Comparisons: 124750
Swaps:       250
Time:        0.00000 ms
Sorted correctly: YES

------------------------------------------
   SORTING ALGORITHM PERFORMANCE ANALYZER
------------------------------------------
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 5

Insertion Sort result:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120
121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140
141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160
161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180
181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200
201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220
221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240
241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260
261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280
281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320
321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340
341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360
361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380
381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400
401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420
421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440
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401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420
421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440
441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460
461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480
481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500

Comparisons: 124750
Shifts:      124750
Time:        0.00000 ms
Sorted correctly: YES

------------------------------------------
   SORTING ALGORITHM PERFORMANCE ANALYZER
------------------------------------------
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 6
Enter number of timing trials (1-100): 1
```

```
Algorithm          | Comparisons  | Swaps/Shifts | Average time (ms)
-------------------+--------------+--------------+-----------------
Selection Sort     | 124750       | 250          | 0.000
Insertion Sort     | 124750       | 124750       | 0.000
```

```
Selection output is sorted: YES
Insertion output is sorted: YES
Both outputs are identical: YES

Interpretation:
- Selection Sort performs n(n-1)/2 comparisons for every arrangement.
- Insertion Sort usually performs fewer comparisons on already sorted data.
- Timing is averaged across the requested trials; very small runs may show 0.000 ms.
- Repeat with different sizes and arrangements to compare practical behaviour.
```

4) Sorted order (ascending)

```
==========================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
==========================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 2
Enter dataset size (1-10000): 10

Choose arrangement:
1. Sorted order (ascending)
2. Reverse-sorted order (descending)
3. Random order
Enter choice: 1
Dataset generated successfully.

==========================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
==========================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 4

Selection Sort result:
1 2 3 4 5 6 7 8 9 10
Comparisons: 45
Swaps:       0
Time:        0.00000 ms
Sorted correctly: YES

==========================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
==========================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
```

```
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 5

Insertion Sort result:
1 2 3 4 5 6 7 8 9 10
Comparisons: 9
Shifts:      0
Time:        0.00000 ms
Sorted correctly: YES

==========================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
==========================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 6
Enter number of timing trials (1-100): 1

---------- PERFORMANCE COMPARISON ----------
Dataset size: 10 | Timing trials: 1
Original dataset:
1 2 3 4 5 6 7 8 9 10

Algorithm          | Comparisons  | Swaps/Shifts | Average time (ms)
-------------------+--------------+--------------+-----------------
Selection Sort     | 45           | 0            | 0.000
Insertion Sort     | 9            | 0            | 0.000

Selection Sort output:
1 2 3 4 5 6 7 8 9 10
Insertion Sort output:
1 2 3 4 5 6 7 8 9 10
Selection output is sorted: YES
Insertion output is sorted: YES
Both outputs are identical: YES
```

```
Interpretation:
- Selection Sort performs n(n-1)/2 comparisons for every arrangement.
- Insertion Sort usually performs fewer comparisons on already sorted data.
- Timing is averaged across the requested trials; very small runs may show 0.000 ms.
- Repeat with different sizes and arrangements to compare practical behaviour.

==========================================
   SORTING ALGORITHM PERFORMANCE ANALYZER
==========================================
1. Enter dataset manually
2. Generate dataset
3. Display original dataset
4. Run Selection Sort
5. Run Insertion Sort
6. Compare both algorithms
7. Exit
------------------------------------------
Enter your choice: 7
Thank you for using the analyzer!
PS C:\Users\anurag chourasiya\Desktop\AOA mini project>
```